

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Shows how this particular value has been calculated As in typical solution Or $(-6.51 - (-6)) \div (3.1 - 3)$	AO1.1b	B1	$(-6.51 - (-6)) \div 0.1$
(b)	Calculates values for $(x + h)$ and $f(x + h)$ CAO	AO1.1b	B1	3.01, -6.0501
	Calculates value for gradient	AO1.1b	B1	-5.01
(c)	Infers suggested limit	AO2.2b	B1	-5
(d)	Recalls and applies formula for gradient	AO1.2	M1	$\begin{aligned} \text{Gradient} &= \frac{f(3+h) - f(3)}{h} \\ &= \frac{(3+h) - (3+h)^2 - (3-3^2)}{h} \\ &= \frac{-h^2 - 5h}{h} \\ &= -h - 5 \end{aligned}$
	Substitutes correct expressions for $f(3+h)$ and $f(3)$	AO1.1b	A1	
	Simplifies to obtain $-h - 5$	AO1.1b	A1	
	Evaluates gradient at $x = 3$ and shows that it is the required value. Constructs rigorous mathematical argument to show the required result. Only award if they have a completely correct solution, using $h \rightarrow 0$ (not $= 0$)	AO2.1	R1	
(d)	Alternative Recalls and applies formula for gradient	AO1.2	M1	$\begin{aligned} \text{Gradient} &= \frac{f(x+h) - f(x)}{h} \\ &= \frac{(x+h) - (x+h)^2 - (x-x^2)}{h} \\ &= \frac{-h^2 - (2x-1)h}{h} \\ &= -h - (2x-1) \end{aligned}$
	Substitutes correct expressions for $f(x+h)$ and $f(x)$	AO1.1b	A1	
	Simplifies to obtain $-h - (2x - 1)$	AO1.1b	A1	

Evaluates gradient at $x = 3$ and shows that it is the required value. Constructs rigorous mathematical argument to show the required result. Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips.	AO2.1	R1	As $h \rightarrow 0$, gradient $\rightarrow -(2x - 1)$ When $x = 3$ gradient = -5
Total 8 marks			

Q2.

Marking Instructions (I)	AO	Marks	Typical Solution (using r)
Obtains correct weld length in terms of h and r	AO1.1b	B1	Length of weld = $w = h + 2\pi r$ Volume = $8000 = \pi r^2 h$
Obtains formula for h in terms of r or vice versa using volume = 8000	AO3.1b	M1	So $h = \frac{8000}{\pi r^2}$
Substitutes to get weld length in terms of one variable, obtaining correct formula for w	AO1.1b	A1	$w = \frac{8000}{\pi r^2} + 2\pi r$
States that for a stationary point the first derivative is zero. (OE)	AO2.4	E1	For minimum length of weld $\frac{dw}{dr} = 0$
Differentiates correctly (FT provided formula includes negative powers) (accept numerical value of $\frac{16000}{\pi}$ used)	AO1.1b	B1F	$\frac{-16000}{\pi r^3} + 2\pi = 0$ leading to $\pi^2 r^3 = 8000$
Solves to find a value of r and a value of h (do not award if the final value of r or h is negative)	AO1.1a	M1	$r = 9.32 \text{ cm}$ $h = 29.3 \text{ cm}$
Obtains $r = 9$ or 9.3 (AWRT) and $h = 29, 29.3$ (AWRT) or 30	AO1.1b	A1	

Differentiates 'their' first differential and substitutes in 'their' value of r or h	AO1.1a	M1	$\frac{d^2w}{dr^2} = \frac{48000}{\pi r^4}$ which is positive, so this is a minimum for w
Sets out a well-constructed mathematical argument, using precise statements throughout to find the values of r (9 or 9.3) and h (29, 29.3 or 30) and justifies the minimum value. Can be awarded if E1 not obtained.	AO2.1	R1	

Marking Instructions (II)	AO	Marks	Typical Solution (using h)
			Length of weld = $w = h + 2\pi r$ Volume = $8000 = \pi r^2 h$ $r = \sqrt{\frac{8000}{\pi h}}$ $w = h + 2\pi \sqrt{\frac{8000}{\pi h}}$ $= h + \sqrt{32000\pi h}^{\frac{1}{2}}$ For minimum length of weld $\frac{dw}{dh} = 0$ $1 - \frac{1}{2} \sqrt{32000\pi h}^{\frac{3}{2}} = 0$ leading to $h^3 = 8000\pi$ $h = 29.3 \text{ cm}$ $r = 9.32 \text{ cm}$ $\frac{d^2w}{dh^2} = \frac{3}{4} \sqrt{32000\pi h}^{\frac{5}{2}}$ which is positive, so this is a minimum for w
Total 9 marks			

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms expression of the correct form for the gradient of the line AB condone sign error	AO1.1a	M1	Gradient of AB $= \frac{(-4+h)^3 - 48(-4+h) - ((-4)^3 - 48(-4))}{h}$
	Obtains correct expansion of $(-4+h)^3$	AO1.1b	B1	$= \frac{h^3 - 12h^2 + 48h - 64 - 48h + 192 - 128}{h}$
	Obtains correct expansion of numerator	AO1.1b	A1	$= \frac{h^3 - 12h^2}{h}$
	Simplifies numerator and shows given result	AO2.1	R1	$= h^2 - 12h$
(b)	Explains that as $h \rightarrow 0$ the gradient of the line $AB \rightarrow$ the gradient of the curve or tangent to the curve Or gradient of curve is given by $\lim_{h \rightarrow 0} h^2 - 12h$ Must not use $h = 0$	AO2.4	E1	The gradient of the curve is given by $\lim_{h \rightarrow 0} h^2 - 12h$
	Explains that $\lim_{h \rightarrow 0} h^2 - 12h = 0$ therefore A must be a stationary point	AO2.4	E1	As $h \rightarrow 0$, $h^2 - 12h \rightarrow 0$ therefore A must be a stationary point
				Total 6 marks

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Deduces that the lower bound of x is 1	AO2.2a	M1	$\{x \in \mathbb{R} : x > 1\}$
	States the domain in a correct form	AO2.5	A1	

(b)	Differentiates using quotient rule (condone correct use of product rule) Must have $f'(x) = \frac{(2x-2)^{\frac{1}{2}} - kx(2x-2)^{-\frac{1}{2}}}{(2x-2)}$ OE	AO1.1a	M1	$f'(x) = \frac{(2x-2)^{\frac{1}{2}} - \frac{1}{2}x(2x-2)^{-\frac{1}{2}} \times 2}{(2x-2)}$ $= \frac{2x-2-x}{(2x-2)^{\frac{3}{2}}}$ $= \frac{x-2}{(2x-2)^{\frac{3}{2}}}$
	Obtains correct derivative in unsimplified form	AO1.1b	A1	
	Completes algebraic manipulation, with all previous working correct, to show the correct form. AG	AO2.1	R1	
(c)	States that point of inflection requires second derivative to be 0	AO2.4	E1	For point of inflection $f''(x) = 0$ $f''(x) = \frac{(2x-2)^{\frac{3}{2}} - \frac{3}{2}(x-2)(2x-2)^{\frac{1}{2}} \times 2}{(2x-2)^3}$ $(2x-2)^{\frac{3}{2}} - 3(x-2)(2x-2)^{\frac{1}{2}} = 0$ $(2x-2)^{\frac{1}{2}}[(2x-2) - 3(x-2)] = 0$ $(2x-2)^{\frac{1}{2}}(4-x) = 0$
	Forms an equation $f''(x) = 0$ OE	AO1.1a	M1	
	Solves their equation	AO1.1a	M1	
	Obtains solution $x = 4$	AO1.1b	A1	
	Gives a valid reason for rejecting $x = 1$, or cancels factor of $(2x-2)^{1/2}$ stating $x \neq 1$.	AO2.4	E1	
	Tests either side of 'their' $x = 4$	AO1.1a	M1	
	Completes rigorous argument to conclude they have one point of inflection Do not award this mark if 2 nd E1 mark not awarded	AO2.1	R1	
(d)	Deduces values of x for convex section of graph	AO2.2a	B1	$x = 1$ or $x = 4$ $x \neq 1$ because of domain $f''(3) = \frac{1}{32} > 0$ $f''(5) = \frac{-\sqrt{2}}{256} < 0$ Therefore point of inflection at $x = 4$
				Total 13 marks

Q5.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Substitutes $3 + h$ to obtain a correct unsimplified expression for $f(3 + h)$	AO1.1a	M1	$(3 + h)^2 - 4(3 + h) + 2$ or $= 9 + 6h + h^2 - 12 - 4h + 2$
	Expresses simplified answer correctly in given format	AO1.1b	A1	$= h^2 + 2h - 1$

(b)	Identifies and uses $\frac{f(x+h)-f(x)}{h}$ to obtain an expression for the gradient of chord Mark can be awarded for unsimplified expression.	AO1.1a	M1	$\text{Gradient of chord} = \frac{f(3+h)-f(3)}{h}$ $= \frac{h^2+2h+1-1}{h}$ $= h+2$
	Obtains a correct and full simplification	AO1.1b	A1	As $h \rightarrow 0$, $h+2 \rightarrow 2$ Gradient of tangent = 2
	Deduces that, as h approaches 0 the limit of $\frac{f(3+h)-f(3)}{h}$ is 2 (Must not simply say $h=0$ but accept words rather than limit notation) FT 'their' gradient provided M1 has been awarded	AO2.2a	R1	
Total 5 marks				

Q6.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Selects an appropriate routine procedure; evidence of quotient rule or product rule	AO1.1a	M1	$\frac{dy}{dx} = \frac{2(4x^2+7)-8x(2x+3)}{(4x^2+7)^2}$
	Obtains correct derivative (no need for simplification)	AO1.1b	A1	
(ii)	States clearly that $\frac{dy}{dx} > 0 \Rightarrow y$ is increasing	AO2.4	R1	$y \text{ is increasing} \Leftrightarrow \frac{dy}{dx} > 0$ $\frac{2(4x^2+7)-8x(2x+3)}{(4x^2+7)^2} > 0$
	Forms inequality from 'their' $\frac{dy}{dx} > 0$	AO3.1a	B1	$(4x^2+7)^2 > 0 \text{ for all } x$ $\therefore 2(4x^2+7)-8x(2x+3) > 0$
	Deduces numerator must be positive	AO2.2a	R1	$8x^2+14-16x^2-24x > 0$
	Considers denominator alone and sets out clear argument to justify given	AO2.1	R1	$4x^2+12x-7 < 0 \text{ (AG)}$

	inequality AG Only award this mark if they have a completely correct solution, which is clear, easy to follow and contains no slips			
(b)	Solves the correct quadratic inequality (accept evidence of factorising, completing the square, use of formula, or correct critical values stated)	AO1.1a	M1	$(2x + 7)(2x - 1)$ $x = -\frac{7}{2}, \frac{1}{2}$
	Obtains fully correct answer, given as an inequality or using set notation	AO1.1b	A1	$-\frac{7}{2} < x < \frac{1}{2}$ $x \in \left(-\frac{7}{2}, \frac{1}{2}\right)$ Or $x \in \left\{x : -\frac{7}{2} < x < \frac{1}{2}\right\}$ Or
Total 8 marks				

Q7.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Identifies and clearly defines variables.	AO3.1b	B1	Let C = total cost
	Models the cost with an expression of the form $ax^2 + bxh$	AO3.3	M1	x = length of base edges h = length of height $C = 15x^2 + 32xh$
	Eliminates either variable, using volume equation, to form a model for the cost in one variable.	AO3.3	M1	$x^2h = 60$ $h = \frac{60}{x^2}$
	Obtains a correct equation to model cost in one variable	AO3.1b	A1	$C = 15x^2 + \frac{1920}{x}$ (*) Differentiating $30x - \frac{1920}{x^2} = 0$
	Uses their model to find minimum. (at least one term correctly differentiated and equated to zero)	AO3.4	M1	$x^3 = 64$ $x = 4$ $h = \frac{60}{4^2} = 3.75$
	Obtains correct equation	AO1.1b	A1	Height of tank is 3.75 m

Obtains correct value for h with correct units in context Award FT from correct substitution into incorrect equation for h but only if all three M1 marks have been awarded, must have correct units.	AO3.2a	A1F	$\frac{d^2C}{dx^2} = 30 + \frac{3840}{x^3}$ $x = 4 \Rightarrow \frac{d^2C}{dx^2} > 0 \text{ therefore minimum}$ ALT from (*) onwards $C = 900h^{-1} + 32\sqrt{60}h^{\frac{1}{2}}$ $0 = -900h^{-2} + 16\sqrt{60}h^{-\frac{1}{2}}$ Height of tank is 3.75 m $\frac{d^2C}{dh^2} = 1800h^{-3} - 8\sqrt{60}h^{-\frac{3}{2}}$ $h = 3.75 \Rightarrow \frac{d^2C}{dh^2} = 25.6 > 0$ $\therefore$ minimum
Performs a correct test of 'their' solution: uses the second derivative of 'their' expression for C in terms of x or h to justify that a minimum value for h has been found OE (Second derivative > 0 or test gradient/values either side)	AO2.4	R1	

(b)	Explains that two of the sides of the tank will need to have length $x \pm 0.05$ in order to join them	AO3.5c	E1	The sides will need to overlap to be joined, so two of the side lengths will need to be $x + 0.05$
(i)				
(ii)	Explains that the refinement is relatively small and unlikely to have a significant effect on the result.	AO3.5a	R1	The minimum cost is likely to increase slightly, but relative to the size of the tank, an extra 5cm is unlikely to make a significant difference.
Total 10 marks				

Q8.

Marking Instructions	AO	Marks	Typical Solution
Translates $f'\left(\frac{\pi}{6}\right)$ into $\lim_{h \rightarrow 0} \frac{\sin\left(\frac{\pi}{6} + h\right) - \sin\left(\frac{\pi}{6}\right)}{h}$	AO1.1a	M1	$f'\left(\frac{\pi}{6}\right) = \lim_{h \rightarrow 0} \left[\frac{\sin\left(\frac{\pi}{6} + h\right) - \sin\left(\frac{\pi}{6}\right)}{h} \right]$ $= \lim_{h \rightarrow 0} \left[\frac{\sin\frac{\pi}{6} \cos h + \cos\frac{\pi}{6} \sin h - \sin\left(\frac{\pi}{6}\right)}{h} \right]$
Uses $\sin(A+B)$ identity to replace	AO2.1	M1	

$\sin\left(\frac{\pi}{6} + h\right)$, to commence argument (at least two lines of argument seen)			$= \lim_{h \rightarrow 0} \left[\frac{\frac{1}{2} \cos h + \frac{\sqrt{3}}{2} \sin h - \frac{1}{2}}{h} \right]$
Obtains correct two term expression involving $\cos h$ and $\sin h$	AO2.5	M1	$= \lim_{h \rightarrow 0} \left[\frac{1}{2} \left(\frac{\cos h - 1}{h} \right) + \frac{\sqrt{3}}{2} \frac{\sin h}{h} \right]$
Deduce what happens as $h \rightarrow 0$, for one part of 'their' expression using the limit of $\frac{\sin h}{h}$ OR by using small angle approximations	AO2.2a	R1	$= \lim_{h \rightarrow 0} \left[\frac{1}{2} \left(\frac{-2 \sin^2\left(\frac{h}{2}\right)}{\frac{2h}{2}} \right) + \frac{\sqrt{3}}{2} \frac{\sin h}{h} \right]$ $= \left(\lim_{h \rightarrow 0} \frac{-\sin\left(\frac{h}{2}\right)}{2} \right) \left(\lim_{h \rightarrow 0} \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)} \right) + \frac{\sqrt{3}}{2} \lim_{h \rightarrow 0} \frac{\sin h}{h}$
Deduce what happens as $h \rightarrow 0$, for the second part of 'their' expression using the limit of $(\cos h - 1)h$ OR by using small angle approximations	AO2.2a	R1	$= 0 \times 1 + \frac{\sqrt{3}}{2} \times 1$ $= \frac{\sqrt{3}}{2}$
Completes a rigorous argument leading to the correct exact value, with all the steps in the method clearly shown.	AO2.1	R1	
			Total 6 marks

Q9.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds 2 nd derivative and sets up an inequality	AO3.1a	M1	$\frac{dy}{dx} = -2xe^{-x^2}$
	Obtains correct first derivative	AO1.1b	A1	$\frac{d^2y}{dx^2} = -2e^{-x^2} + 4x^2e^{-x^2}$
	Obtains second derivative correct from 'their' first derivative	AO1.1b	A1F	$-2e^{-x^2} + 4x^2e^{-x^2} < 0$ $4x^2 - 2 < 0$
	Deduces correct final inequality (could use set	AO2.2a	A1	$-\frac{\sqrt{2}}{2} < x < \frac{\sqrt{2}}{2}$

	notation)			
(b)	Uses trapezium rule	AO1.1a	M1	$\int_{0.1}^{0.5} e^{-x^2} dx \approx \frac{0.1}{2} (e^{-0.01} + e^{-0.25})$ $+ 2(e^{-0.04} + e^{-0.09} + e^{-0.16})$ ≈ 0.3611
	Trapezium rule entries all correct	AO1.1b	A1	
	Finds correct value	AO1.1b	A1	
(c)	References area being completely within concave section So...	AO2.4	E1	$[0.1, 0.5] \subset \left[-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]$ $\therefore$ area is completely within concave section
	Trapezia all fall completely underneath the curve therefore underestimate (only award this mark if previous E1 has been awarded)	AO2.4	E1	Hence trapezia lie below curve and give an under-estimate for the area
(d)	Uses suitable rectangle to obtain over-estimate	AO3.1a	B1	Using a rectangle with the left hand edge the same height as the curve will produce an over-estimate Area of rectangle = $0.4 \times e^{-0.01^2} = 0.396...$ $\therefore 0.36 < A < 0.40$ So $A = 0.4$ to 1 dp
	Explains that this rectangle lies above the curve	AO2.4	E1	
	Constructs rigorous mathematical argument about accuracy, which leads to required result Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips.	AO2.1	R1	
				Total 12 marks

Q10.

(a) $\left(\frac{dy}{dt}\right) = \frac{4t^3}{8} - 2t$

one of these terms correct

M1

all correct (no + c etc)

A1

(b) (i) $t = 1 \Rightarrow \frac{dy}{dt} = \frac{4}{8} - 2$

Correctly sub $t = 1$ into their $\frac{dy}{dt}$

M1

$$= -1\frac{1}{2}$$

must have $\frac{dy}{dt}$ correct (watch for t^3 etc)

A1cso

(ii) $\frac{dy}{dt} < 0$

must have used $\frac{dy}{dt}$ in part (b)(i)

$\Rightarrow$ (height is) **decreasing** (when $t = 1$)

must state that “ $\frac{dy}{dt} < 0$ ” or “ $-1.5 < 0$ ” or the equivalent in words

FT their value of $\frac{dy}{dt}$ with appropriate explanation and conclusion

E1✓

(c) (i) $\left(\frac{d^2y}{dt^2} = \right) \frac{4}{8} \times 3t^2 - 2$

Correctly differentiating their $\frac{dy}{dt}$ even if wrongly simplified

M1

$\left(t = 2, \frac{d^2y}{dt^2} = \right) 4$

Both derivatives correct and simplified to 4

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A1cso

2

(ii) $\Rightarrow$ minimum

FT their numerical value of $\frac{d^2y}{dt^2}$ from part (c)(i)

E1✓

1

[8]

Q11.

(a) (Gradient = $10 - 6 + 5$) = 9

correct gradient from sub $x = 1$ into $\frac{dy}{dx}$

B1

$y - 4 = \text{"their 9"} (x - 1)$
 or $y = \text{"their 9"} x + c$ **and** attempt
 to find c using $x = 1$ and $y = 4$

}

*must attempt to use given expression for $\frac{dy}{dx}$ **and** must
 be attempting tangent and not normal **and** coordinates
 must be correct*

M1

$$y = 9x - 5$$

condone $y = 9x + c, \dots \quad c = -5$

A1

3

(b) $(y =) \frac{10}{5}x^3 - \frac{6}{3}x^3 + 5x + C$
one term correct

another term correct

M1

A1

all integration correct including $+ C$

A1

$$4 = 2 - 2 + 5 + C \Rightarrow C = -1$$

*substituting both $x = 1$ and $y = 4$ **and** attempting
 to find C*

m1

$$y = 2x^5 - 2x^3 + 5x - 1$$

must have $y = \dots$ and coefficients simplified

A1cso

5

[8]

Q12.

(a) $f(-3) = (-3)^3 - 4(-3) + 15$
 $f(-3)$ attempted not long division

M1

$$f(-3) = -27 + 12 + 15 = 0 \Rightarrow x + 3 \text{ is a factor}$$

shown = 0 plus statement

A1

2

- (ii) Quadratic factor ($x^2 - 3x + 5$)
-3x or + 5 term by inspection or full long division attempt

$$(f(x) =) (x + 3)(x^2 - 3x + 5)$$

must see correct product

A1

2

(b) (i) $\left(\frac{dy}{dx} =\right) 4x^3 - 16x + 60$
one of these terms correct

another term correct

all correct (no +c etc)

M1

A1

A1

3

(ii) $4x^3 - 16x + 60 = 0$
must see this line OE

$$\Rightarrow x^3 - 4x + 15 = 0$$

AG

B1

1

- (iii) Discriminant of quadratic = $(-3)^2 - 4 \times 5$
discriminant of "their" quadratic or correct use of quad eqn "formula"

M1

$$b^2 - 4ac = -11 \text{ (or } b^2 - 4ac < 0 \text{)}$$

therefore quadratic has no (real) roots

Hence only stationary point is when $x = -3$

correct discriminant evaluated correctly
(or shown to be < 0) with appropriate conclusion
plus final statement

A1

2

(iv) $\left(\frac{d^2y}{dx^2} = \right) 12x^2 - 16$

B1 ✓

$= 12(-3)^2 - 16$ (or $12 \times 9 - 16$ etc)

sub $x = -3$ into "their" $\frac{d^2y}{dx^2}$

M1

$= 92$

A1

3

- (v) Minimum **since** $\frac{d^2y}{dx^2} > 0$ (or $92 > 0$ etc)
FT appropriate conclusion from their value from
(iv) **plus** reason
treat parts (iv) & (v) holistically

E1 ✓

1

[14]

Q13.

(a) (i) $\left(\frac{dy}{dx} = \right) 5x^4 - 6x + 1$
one term correct

M1

another term correct

A1

all correct (no $+ c$ etc)

A1

(ii) $\left(\frac{d^2y}{dx^2} = \right) 20x^3 - 6$
FT 'their' $\frac{dy}{dx}$

B1✓

(b) $x = -1 \Rightarrow \frac{dy}{dx} = 5(-)^4 - 6(-1) + 1 (= 12)$
must sub $x = -1$ into 'their' $\frac{dy}{dx}$

M1

$\Rightarrow y = 12(x + 1)$
any correct form with $(x - -1)$ simplified
condone $y = 12x + c$, $c=12$

A1cso

(c) $x = 1 \Rightarrow \frac{dy}{dx} = 5 - 6 + 1$
sub $x = 1$ into their $\frac{dy}{dx}$

$\frac{dy}{dx} = 0 \Rightarrow$ stationary point
shown = 0 plus correct statement

A1cso

when $x = 1$, $\frac{d^2y}{dx^2} = 14 \Rightarrow (B \text{ is a })$ minimum (point)
 or $\frac{d^2y}{dx^2} = 20 - 6 > 0 \Rightarrow (B \text{ is a })$ minimum (point)
must have correct $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for E1

E1

(d) (i) $\frac{x^6}{6} - \frac{3x^3}{3} + \frac{x^2}{2} + 5x$
one term correct

M1

another term correct

A1

all correct (may have + c)

A1

$\left[\frac{1}{6} - 1 + \frac{1}{2} + 5 \right] - \left[\frac{1}{6} + 1 + \frac{1}{2} - 5 \right]$
'their' $F(1) - F(-1)$ with powers of 1 and

-1 evaluated correctly

M1

$$= 8$$

A1cso

5

(ii) 'their answer to part (i)' - 2

M1

$$\Rightarrow \text{Area} = 6$$

A1cso

2

[16]

Q14.

- (a) (i) $3x^2 + 3x^2 + xy + xy + 3xy + 3xy$
correct expression for surface area

M1

$$6x^2 + 8xy = 32 \Rightarrow 3x^2 + 4xy = 16$$

$$2(3x^2 + xy + 3xy) = 32 \text{ etc}$$

AG be convinced

A1

2

- (ii) $(V =) 3x^2 y$ OE
correct volume in terms of x and y

M1

$$= 3x \left(\frac{16 - 3x^2}{4} \right) \text{ or } = 3x^2 \left(\frac{16 - 3x^2}{4x} \right)$$

OE

$$= 12x - \frac{9x^3}{4}$$

CSO AG

be convinced that all working is correct

A1

2

(b) $\left(\frac{dV}{dx} =\right) 12 - \frac{27}{4}x^2$

one of these terms correct

M1

all correct with 9×3 evaluated (no + c etc)

A1

2

(c) (i) $x = \frac{4}{3} \Rightarrow \frac{dV}{dx} = 12 - \frac{27}{4} \times \left(\frac{4}{3}\right)^2$

attempt to sub $x = \frac{4}{3}$ into 'their' $\frac{dV}{dx}$

M1

$$\frac{dV}{dx} = 12 - \frac{27}{4} \times \frac{16}{9} = 12 - 12$$

$$\text{or } 12 - \frac{432}{36} = 12 - 12 \text{ or } 12 - \frac{48}{4} = 0 \text{ etc}$$

$$\frac{dV}{dx} = 0 \Rightarrow \text{stationary value}$$

CSO; shown = 0 plus statement

A1

2

(ii) $\frac{d^2V}{dx^2} = -\frac{27x}{2}$ OE

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FT for 'their' $\frac{dV}{dx} = a + bx^2$

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when $x = \frac{4}{3}$, $\frac{d^2V}{dx^2} < 0 \Rightarrow \text{maximum}$

or sub of $x = \frac{4}{3}$ into 'their' $\frac{d^2V}{dx^2} \Rightarrow \text{maximum}$

$$\left(\text{FT "minimum" if their } \frac{d^2V}{dx^2} > 0 \right)$$

EO if numerical error seen

E1 ✓

2

[10]

Q15.

(a) (i)

$$\left. \begin{array}{l} (\text{Increasing} \Rightarrow) \frac{dy}{dx} > 0 \\ 20x - 6x^2 - 16 > 0 \end{array} \right\} \text{ either}$$

correct interpretation of y increasing

M1

$$\Rightarrow 6x^2 - 20x + 16 < 0 \quad \text{or} \quad (2) \quad (10x - 3x^2 - 8) > 0$$

must see at least one of these steps before final answer for A1

$$\Rightarrow 3x^2 - 10x + 8 < 0$$

CSO AG no errors in working

A1

2

(ii) $(3x - 4)(x - 2)$

correct factors or correct use of quadratic equation

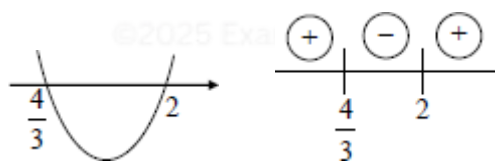
formula as far as $\frac{10 \pm \sqrt{4}}{6}$

M1

CVs are $\frac{4}{3}$ and 2

condone $\frac{8}{6}$ and $\frac{12}{6}$ here but not in final line

A1



sketch or sign diagram

M1

$$\frac{4}{3} < x < 2$$

$$\text{or } 2 > x > \frac{4}{3}$$

$$\text{accept } x < 2 \text{ AND } x > \frac{4}{3}$$

$$\text{but not } x < 2 \text{ OR } x > \frac{4}{3}$$

$$\text{nor } x < 2, x > \frac{4}{3}$$

A1

Mark their final line as their answer

4

(b) (i) $x = 2 ; \left(\frac{dy}{dx} = \right) 40 - 24 - 16$

sub $x = 2$ into $\frac{dy}{dx}$ and simplify terms

M1

$\frac{dy}{dx} = 0 \Rightarrow$ tangent at P is parallel to the x -axis
must be all correct working plus statement

A1

2

(ii) $x = 3 ; \frac{dy}{dx} = 20 \times 3 - 6 \times 3^2 - 16$

must attempt to sub $x = 3$ into $\frac{dy}{dx}$

M1

$$(= 60 - 54 - 16) = -10$$

A1

Gradient of normal = $\frac{1}{10}$

$$\frac{-1}{\text{"their"} -10}$$

A1 ✓

Normal: $(y - -1) = \text{'theirgrad'}(x - 3)$

normal attempted with correct coordinates

used and gradient obtained from their $\frac{dy}{dx}$ value

m1

$$y + 1 = \frac{1}{10}(x - 3)$$

any correct form, eg $10y = x - 13$ but
must simplify -- to +

A1

(Equation of tangent at P is) $y = 3$

B1

$$x = 43$$

$$CSO; \Rightarrow R(43, 3)$$

A1

7

[15]

Q16.

- (a) (i) (When $x = 2$) $\frac{dy}{dx} = 12 - 1 - 11 = 0$
AG Must see intermediate evaluations

B1

1

- (ii) $\frac{4}{x^2} = 4x^{-2}$ {so $\frac{dy}{dx} = 3x^2 - 4x^{-2} - 11$ }
 $\frac{4}{x^2} = 4x^{-2}$, seen in (a)(ii) or earlier. PI by $\pm 8x^{-3}$ term in answer

B1

$$\frac{d^2y}{dx^2} = 6x + 8x^{-3}$$

Correct powers of x correctly obtained from differentiating the first two terms

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M1

$$6x + 8x^{-3} \text{ ACF}$$

A1

When $x = 2$, $\frac{d^2y}{dx^2} = 12 + 8/8 = 13$

A1

4

- (iii) Since $\frac{d^2y}{dx^2} > 0$, P is a minimum point.
Ft on c 's value of $y''(2)$ in (a)(ii) but must see reference to sign of $y''(2)$ either explicitly or as inequality, as well as the correct ft conclusion

E1F

1

(b) $\int \left(3x^2 - \frac{4}{x^2} - 11 \right) dx = x^3 + 4x^{-1} - 11x (+c)$

Attempt to integrate $\frac{dy}{dx}$ with at least two of the three terms integrated correctly

M1

$(y =) x^3 + 4x^{-1} - 11x(+c)$

For $x^3 + 4x^{-1} - 11x$ OE even unsimplified.

1

When $x = 2, y = 1 \Rightarrow 1 = 8 + 2 - 22 + c$

Substituting

$x = 2, y = 1$ into $y = F(x) + 'c'$ in attempt to find constant of integration, where $F(x)$ follows attempted integration of

expression for $\frac{dy}{dx}$

$y = x^3 + 4x^{-1} - 11x + 13$

ACF

A1

4

[10]

Q17.

(a) $\left(\frac{dx}{dy} = \right) \frac{\cos y \times \cos y - \sin y \times -\sin y}{\cos^2 y}$

Condone incorrect signs, poor notation, omission of

$\frac{dy}{dx}$ or using $\frac{dx}{dy}$

M1

$= \frac{\cos^2 y + \sin^2 y}{\cos^2 y}$

*RHS correct with terms squared, including correct notation
Must see this line*

A1

$= \frac{1}{\cos^2 y}$

or $(= 1 + \tan^2 y)$

Must see one of these

$$\frac{dx}{dy} = \sec^2 y$$

AG; all correct including correct use of $\frac{dx}{dy}$ throughout

A1
CSO

3

(b) $\sec^2 y = 1 + (x - 1)^2$

Correct use of $\sec^2 y = 1 + \tan^2 y$ and in terms of x

M1

$$= 1 + x^2 - 2x + 1 \quad \text{OE}$$

$$= x^2 - 2x + 2$$

AG; must see " $\sec^2 y =$ ", $(x - 1)^2$ expanded and no errors seen

A1

2

(c) $\frac{dx}{dy} = x^2 + 2x + 2$ or $\frac{dy}{dx} = \frac{1}{\sec^2 y}$

Must be seen

$$\frac{dy}{dx} = \frac{1}{x^2 - 2x + 2}$$

AG and no errors seen

B1

1

(d) (i) $y = \tan^{-1}(x - 1) - \ln x$

$$\left(\frac{dy}{dx} = \right) \frac{1}{x^2 - 2x + 2} - \frac{1}{x}$$

Must be correct

M1

$$\left(\frac{dy}{dx} = 0 \right)$$

$$\pm x^2 + bx + c (= 0)$$

Expression in this form (generous), where b and $c \neq 0$

m1

$$x^2 - 3x + 2 = 0$$

Must see correct equation = 0

A1

$$x = 1, 2$$

Both answers must be seen

The two A marks are independent

A1

4

(ii) $y'' = p(x^2 - 2x + 2)^{-2}(2x - 2) \pm qx^{-2}$
where p and q are constants

M1

$$y'' = -(x^2 - 2x + 2)^{-2}(2x - 2) + x^{-2}$$

p = -1, q = 1 including correct brackets

A1

2

(iii) $x = 1, y'' = 1$

Must have scored full marks in (d)(i) and (ii)

M1

At $x = 1, y'' > 0 \therefore \text{min}$

When $x = 1, y = 0$ hence on x -axis

Must see $y'' > 0$ or in words

Both statements fully correct

A1

2

EXAM PAPERS PRACTICE

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[14]

Q18.

(a) $y = x^4 + x$
 $\{y(-2 + h) = \} (-2 + h)^4 + (-2 + h)$
 $(-2 + h)^4 + (-2 + h) \text{ PI}$

M1

$$= h^4 - 8h^3 + 24h^2 - 32h + 16 - 2 + h$$

Correct expansion of $(-2 + h)^4$ as

$$h^4 - 8h^3 + 24h^2 - 32h + 16 \text{ PI}$$

B1

$$= h^4 - 8h^3 + 24h^2 - 31h + 14$$

Seen separately or as part of the gradient expression.
Ft one incorrect term in expansion of $(-2 + h)^4$

A1F

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{h^4 - 8h^3 + 24h^2 - 31h + 14 - (14)}{-2 + h - (-2)}$$

Use of correct formula for gradient PI

M1

$$= \frac{h^4 - 8h^3 + 24h^2 - 31h}{h} = h^3 - 8h^2 + 24h - 31$$

The four correct terms in any order A0 if incorrect
(constant / h) term ignored due printed form of answer

A1

5

- (b) As $h \rightarrow 0$, gradient of line in (a) $\rightarrow$ gradient of curve at point $(-2, 14)$

$\text{Lim } [c's(p + qh + rh^2 + h^3)]$ OE

$h \rightarrow 0$

NB ' $h = 0$ ' instead of ' $h \rightarrow 0$ ' gets E0

E1

{Gradient of curve at point $(-2, 14)$ is} -31

Dependent on previous E1 and printed **form** of answer
in (a) obtained convincingly but then ft on c's p value

E1

2

EXAM PAPERS PRACTICE

[7]

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Q19.

(a) $\frac{dy}{dx} = 18 + 6x - 12x^2$

one of these terms correct

M1

another term correct

A1

all correct (no + c etc)
(penalise + c once only in question)

A1

3

(b) $18 + 6x - 12x^2 = 0$

putting their $\frac{dy}{dx} = 0$,

PI by attempt to solve or factorise

M1

$6(3 - 2x)(x + 1) (= 0)$

attempt at factors of **their quadratic**
or use of quadratic equation formula

m1

$x = -1, x = \frac{3}{2}$ OE

must see both values unless $x = -1$
is verified separately

A1

3

If M1 not scored, award SC B1 for verifying

that $x = -1$ leads to $\frac{dy}{dx} = 0$ and a further

SC B2 for finding $x = \frac{3}{2}$ as other value

(c) (i) $\frac{d^2y}{dx^2} = 6 - 24x$

FT their $\frac{dy}{dx}$ but $\frac{d^2y}{dx^2}$ must be correct if

3 marks earned in part (a)

B1 ✓

When $x = -1$, $\frac{d^2y}{dx^2} = 6 - (24 \times -1)$

Sub $x = -1$ into their $\frac{d^2y}{dx^2}$

M1

$\frac{d^2y}{dx^2} = 30$

A1cso

3

(ii) Minimum point

must have a value in (c)(i)

FT "maximum" if their value of $\frac{d^2y}{dx^2} < 0$

E1✓

1

[10]

Q20.

(a) $\left(\frac{dV}{dt}\right) = \frac{3t^2}{4} - 3$

one of these terms correct

M1

all correct (no + c etc)

A1

2

(b) (i) $t = 1 \Rightarrow \frac{dV}{dt} = \frac{3}{4} - 3$

substituting $t = 1$ into their $\frac{dV}{dt}$

M1

$= -2\frac{1}{4}$

(- 2.25 OE) BUT must have $\frac{dV}{dt}$ correct

A1cso

2

(ii) Volume is decreasing when $t = 1$

must have used $\frac{dV}{dt}$ in (b)(i) or starts again

because $\frac{dV}{dt} < 0$

must state that $\frac{dV}{dt} < 0$ (or $-2\frac{1}{4} < 0$ etc)

ft increasing plus explanation if their $\frac{dV}{dt} > 0$

E1 ✓

1

(c) (i) $\left(\frac{dV}{dt} = 0 \Rightarrow\right) \frac{3t^2}{4} - 3 = 0$

PI by "correct" equation being solved

M1

$$\Rightarrow t^2 = 4$$

obtaining $t = k$ correctly from their $\frac{dV}{dt}$

A1 ✓

$$t = 2$$

withhold if answer left as $t = \pm 2$

A1cso

3

(ii) $\left(\frac{d^2V}{dt^2} =\right) \frac{3t}{2}$

(condone unsimplified) ft their $\frac{dV}{dt}$

B1 ✓

When $t = 2$, $\frac{d^2V}{dt^2} = 3$ or $\frac{d^2V}{dt^2} > 0$

ft their $\frac{d^2V}{dt^2}$ and value of t from (c)(i)

M1

$\Rightarrow$ minimum

A1cso

3

[11]

Q21.

(a) $(5 + h)^3 = 125 + 75h + 15h^2 + h^3$

Accept unsimplified coefficients

B1

1

- (b) (i) $y(5 + h) = 100 + 65h + 14h^2 + h^3$
PI; ft numerical error in (a)

B1F

Use of correct formula for gradient

M1

Gradient is $65 + 14h + h^2$

A1 if one numerical error made;

ft numerical error already penalised

A2,1F

4

- (ii) As $h \rightarrow 0$ this $\rightarrow 65$

E1 for 'h = 0';

ft wrong values for p, q, r

E2,1F

2

[7]

Q22.

- (a) (i) $\frac{dy}{dt} = \frac{4t^3}{8} - 4t + 4$
one term correct

M1

another term correct

A1

all correct (no + c etc) unsimplified

A1

3

- (ii) $\frac{d^2y}{dt^2} = \frac{12t^2}{8} - 4$
ft one term "correct"

M1

correct unsimplified (penalise inclusion of +c once only in question)

A1

2

(b) $t = 2; \frac{dy}{dt} = 4 - 8 + 4$

Substitute $t = 2$ into their $\frac{dy}{dt}$

M1

$\frac{dy}{dt} = 0 \Rightarrow$ stationary value

CSO; shown = 0 plus statement

A1

$t = 2; \frac{d^2y}{dt^2} = 6 - 4 = 2$

Sub $t = 2$ into their $\frac{d^2y}{dt^2}$

M1

$\Rightarrow y$ has MINIMUM value
CSO

A1

4

(c) (i) $t = 1; \frac{dy}{dt} = \frac{1}{2} - 4 + 4$

Substitute $t = 1$ into their $\frac{dy}{dt}$

M1

$= \frac{1}{2}$

OE; CSO

NMS full marks if $\frac{dy}{dt}$ correct

A1

2

(ii) $\frac{dy}{dt} > 0 \Rightarrow$ (depth is) INCREASING

allow decreasing if states that their $\frac{dy}{dt}$

Reason must be given not just the word increasing or decreasing

E1ft

1

[12]

Q23.

(a)
$$\left(\frac{dy}{dx}\right) = \frac{\cos 4x \cdot 4 \cos 4x - \sin 4x \cdot -4 \sin 4x}{\cos^2 4x}$$

$$\frac{\pm A \cos^2 4x \pm B \sin^2 4x}{\cos^2 4x}$$

M1

$$= \frac{4 \cos^2 4x + 4 \sin^2 4x}{\cos^2 4x} \text{ or better}$$

Both terms correct

A1

$$= 4(1 + \tan^2 4x)$$

CSO

All correct

A1

or

$$\left(\frac{dy}{dx}\right) = \frac{\cos 4x \cdot 4 \cos 4x - \sin 4x \cdot -4 \sin 4x}{\cos^2 4x}$$

$$\frac{\pm A \cos^2 4x \pm B \sin^2 4x}{\cos^2 4x}$$

(M1)

$$= \frac{4 \cos 4x \cos 4x}{\cos 4x \cos 4x} + \frac{4 \sin 4x \sin 4x}{\cos 4x \cos 4x} \text{ or better}$$

(A1)

$$= 4(1 + \tan^2 4x)$$

CSO

All correct

(A1)

3

(b)
$$\frac{d^2 y}{dx^2} = 4 \times 2 \tan 4x \times \dots\dots$$

$$A \tan 4x \times f(4x)$$

M1

$$4 \sec^2 4x$$

$$f(4x) = B \sec^2 4x$$

m1

$$= 32 \tan 4x \sec^2 4x$$

ft 8 × their p from part (a)

A1F

$$= 32 \tan 4x (1 + \tan^2 4x)$$

Previous two method marks must have been earned

m1

$$= 32y(1 + y^2)$$

CSO

A1

Alternative Solutions

$$y' = 4 + 4 \tan^2 4x = 4 + 4 \frac{\sin^2 4x}{\cos^2 4x}$$

$$y' = 4 \times \left[\frac{\cos^2 4x \cdot 2 \sin 4x \cdot 4 \cos 4x + \sin^2 4x \cdot 2 \cos 4x \cdot 4 \sin 4x}{\cos^4 4x} \right]$$

$$\frac{A \cos^3 4x \pm B \sin^3 4x}{\cos^4 4x} \quad \text{where } A \text{ and } B \text{ are constants or trig functions.}$$

(M1)

Where A is m sin 4x and B is n cos 4x

(m1)

$$= \frac{4 \times 8 \sin 4x \cos 4x [\cos^2 4x + \sin^2 4x]}{\cos^4 4x}$$

ft 8 × their p from part (a)

(A1F)

$$= 32 \tan 4x \sec^2 4x$$

$$k \tan 4x \sec^2 4x$$

(m1)

$$= 32y(1 + y^2)$$

CSO

(A1)

or

$$\frac{dy}{dx} = 4 \sec^2 4x$$

$$\frac{d^2 y}{dx^2} = 4 \times 2 \sec 4x \cdot 4 \sec 4x \tan 4x$$

$A \sec 4x \times f(4x)$

(M1)

$$f(4x) = B \sec 4x \tan 4x$$

(m1)

$$= 32 \sec^2 4x \tan 4x$$

ft 8 × their p from part (a)

(A1F)

$$= 32(1 + \tan^2 4x) \tan 4x$$

Previous two method marks must have been earned

(m1)

$$= 32y(1 + y^2)$$

CSO

(A1)

or

$$\frac{dy}{dx} = 4(1 + \tan^2 4x)$$

$$u = \tan 4x \quad \frac{dy}{dx} = 4 + 4u^2$$

$$\frac{d^2 y}{dx^2} = (8)u \frac{du}{dx}$$

(M1)

$$\frac{du}{dx} = 4 + 4 \tan^2 4x = 4 + 4u^2$$

(m1)

$$\frac{d^2 y}{dx^2} = 8u(4 + 4u^2)$$

(A1)

$$= 32u(1 + u^2)$$

(m1)

$$= 32y(1 + y^2)$$

(A1)

5

[8]

Q24.

(a) (i) $t = 0 \quad h = A(1 - 1) = 0$

B1

1

(ii) $57 = A \left(1 - e^{-\frac{12}{4}} \right)$

M1

$$A = \frac{57}{(1 - e^{-1})} \approx 60$$

Or 59.9... seen.

A = correct expression ≈ 60 2 sf

A1

2

(b) (i) $h = 48 \quad \frac{48}{60} = 1 - e^{-\frac{1}{4}t}$

M1

$$\ln \left(e^{-\frac{1}{4}t} \right) = \ln \left(\frac{1}{5} \right)$$

m1

$$-\frac{1}{4}t = -\ln 5 \Rightarrow t = 4 \ln 5$$

A1

3

(ii) $\frac{dh}{dt} = -\frac{1}{4} \times -60 \times e^{-\frac{1}{4}t}$

Differentiate, condone sign errors

M1

$$60e^{-\frac{1}{4}t} = 60 - h \Rightarrow \frac{dh}{dt} = \frac{1}{4}(60 - h)$$

Eliminate $e^{-\frac{1}{4}t}$

m1

$$\frac{dh}{dt} = 15 - \frac{h}{4}$$

CSO, AG

A1

3

(iii) $h = 8$

B1

1

[10]

Q25.

(a) (i) S.A. = $4xy + 5xy + 3xy + 6x^2$ OE
Condone one slip or omission

M1

$$= 12xy + 12x^2$$

A1

$$144 = 12xy + 12x^2$$

Must see this line

$$\Rightarrow xy + x^2 = 12$$

AG

A1cso

(ii) (Volume =) $\frac{1}{2} \times 3x \times 4x \times y$ OE

M1

$$= 6x^2 \times \frac{(12 - x^2)}{x}$$

Must see $(y =) \frac{(12 - x^2)}{x}$ or $xy = 12 - x^2$
for A1

$(V =) 72x - 6x^3$

AG must be convinced not working back
from answer

A1

2

(b) (i) $\frac{dV}{dx} = 72 - 18x^2$
One term correct

M1

All correct (no + c etc)

A1

2

(ii) $x = 2 \Rightarrow \frac{dV}{dx} = 72 - 18 \times 2^2$
 $\Rightarrow \frac{dV}{dx} = 72 - 72 = 0$

Substitute $x = 2$ into their $\frac{dV}{dx}$

M1

$\Rightarrow$ stationary (value when $x = 2$)

Shown = 0 plus statement
Statement may appear first

A1

2

(c) $\frac{d^2V}{dx^2} = -36x$

FT their $\frac{dV}{dx}$

B1ft

$$\frac{d^2V}{dx^2} = -72 \text{ or when } x = 2 \Rightarrow \frac{d^2V}{dx^2} < 0 \Rightarrow \text{maximum}$$

FT their $\frac{d^2V}{dx^2}$ value when $x = 2$
with appropriate conclusion

E1ft

2

[11]

Q26.

(a) $\sqrt{x} = x^{\frac{1}{2}}$

PI

B1

$$\frac{x^3 + \sqrt{x}}{x} = \frac{x^3}{x} + \frac{\sqrt{x}}{x} = x^2 + x^{-\frac{1}{2}}$$

Accept $p = 2; q = -\frac{1}{2}$

B1;B1

3

(b) (i) $\frac{dy}{dx} = 2x - \frac{1}{2}x^{-\frac{3}{2}}$

Reduces both powers by 1

M1

ACF

A1

2

(ii) When $x = 1, y = 2$

PI if not stated explicitly eg the '2'
may appear in the correct posn. in later eqn.

B1

When $x = 1$, $\frac{dy}{dx} = 2 - \frac{1}{2} = \frac{3}{2}$

$\frac{dy}{dx}$
 Attempt to find $\frac{dy}{dx}$ When $x = 1$ PI

M1

Gradient of normal = $-\frac{2}{3}$

$-1/$ (c's value of dy/dx when $x = 1$)
 either stated as the gradient of the
 normal or used as the gradient in the
 equation of the normal

m1

Equation of normal: $y - 2 = -\frac{2}{3}(x - 1)$

$\frac{dy}{dx}$
 Only ft on c's $\frac{dy}{dx}$ in part (b)(i)
 ACftF

A1F

4

(c) (i) $\frac{d^2y}{dx^2} = 2 + \frac{3}{4}x^{-\frac{5}{2}}$

Reduces both powers by 1.

M1

Ft on (b)(i) provided at least one
 power to be differentiated is both
 negative and fractional

A1F

2

(ii) (Since $x > 0$,) $\frac{d^2y}{dx^2} > 0$

$\frac{d^2y}{dx^2}$
 For a maximum point $\frac{d^2y}{dx^2}$ is **not**
 positive so C has no maximum points

$\frac{d^2y}{dx^2}$
 E1 for attempt to find the sign of $\frac{d^2y}{dx^2}$;
 either in general terms or at the pt(s)
 where c's $dy/dx = 0$ or the remaining E mark

a correct justification for why $\frac{d^2y}{dx^2} > 0$ and also
a full correct concluding statement must be made.

E2,1,0

2

[13]

Q27.

(a) At B, $y = (2 + h)^3 - 12(2 + h)$

with attempt to expand and simplify

M1

$$= (8 + 12h + 6h^2 + h^3) - (24 + 12h) = -16 + 6h^2 + h^3$$

correct expansion of $(2 + h)^3$

B1

$$\text{Grad } AB = \frac{(-16 + 6h^2 + h^3) - (-16)}{(2 + h) - 2}$$

m1

$$= \frac{6h^2 + h^3}{h} = 6h + h^2$$

convincingly shown (AG)

A1

- (b) As $h \rightarrow 0$ this gradient $\rightarrow 0$
so gradient of curve at A is 0

E1 for 'h = 0'

E2,1

2

[6]

Q28.

(a) (i) $\frac{dx}{dt} = 2t^3 - 40t + 66$

one term correct

M1

another term correct

A1

all correct unsimplified (no + c etc)

A1

3

(ii) $\frac{d^2x}{dt^2} = 6t^2 - 40$

ft one term correct

M1

ft all "correct", 2 terms equivalent

A1ft

2

(b) $\frac{dx}{dt} = 54 - 120 + 66$

substitute $t = 3$ into their $\frac{dx}{dt}$

M1

$= 0 \Rightarrow$ stationary value

CSO

shown = 0 (54 or 2×27 seen) and statement

A1

Substitute $t = 3$ into $\frac{d^2x}{dt^2} (= 14)$

M1

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$\frac{d^2x}{dt^2} > 0 \Rightarrow$ minimum value

CSO; all values (if stated) must be correct

A1

4

(c) Substitute $t = 1$ into their $\frac{dx}{dt}$

must be their $\frac{dx}{dt}$ NOT $\frac{d^2x}{dt^2}$ etc

M1

$$\frac{dx}{dt} = 28$$

ft their $\frac{dx}{dt}$ when $t = 1$

A1ft

2

(d) Substitute $t = 2$ into their $\frac{dx}{dt}$

must be their $\frac{dx}{dt}$ NOT $\frac{d^2x}{dt^2}$ or x

Interpreting their value of $\frac{dx}{dt}$

M1

$$= 16 - 80 + 66 = 2 \quad (> 0)$$

$\Rightarrow$ increasing when $t = 2$

Allow decreasing if their $\frac{dx}{dt} < 0$

E1ft

2

[13]

Q29.

(a) $y = e^{2x}(x^2 - 4x - 2)$

$$\frac{dy}{dx} = e^{2x}(2x - 4)$$

Product rule; allow 1 slip

M1

$$+ (x^2 - 4x - 2)2e^{2x}$$

A1

$$\frac{dy}{dx} = e^{2x}(2x - 4 + 2x^2 - 8x - 4)$$

Factorising $e^{2x}(2x^2 - 6x - 8)$

M1

$$e^{2x}(2x^2 - 6x - 8)$$

Or $x^2 - 3x - 4 = 0$

A1

$e^{2x} \neq 0$

$(x - 4)(x + 1) = 0$

*Solving 3 term quadratic
Dependent on both M marks*

m1

$x = 4, -1$

And no extras eg $x = 0$

A1

6

(b) (i) $\frac{d^2y}{dx^2} = e^{2x} \cdot 2 + (2x - 4)2e^{2x}$

*Product rule from their $\frac{dy}{dx}$ in form
 e^{2x} (quadratic) $e^{2x}(4x^2 - 8x - 22)$*

M1

$+ (x^2 - 4x - 2)4e^{2x} + 2e^{2x}(2x - 4)$

A1

OR

$\frac{d^2y}{dx^2} = e^{2x}(4x - 6) + (2x^2 - 6x - 8)2e^{2x}$

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M1A1

2

(ii) $x = 4 : y'' = e^8 (10) > 0 \therefore \text{MIN}$

*Their 2 x's in their $\frac{d^2y}{dx^2}$
only of form e^{2x} (quadratic)*

M1

$x = -1 : y'' = e^{-2} (-10) < 0 \therefore \text{MAX}$

*CSO Both correct
Allow values either side of y or y'*

A1

2

Q30.

(a) $\frac{dy}{dx} = 5x^4 + 40x$

One of these powers correct

M1

One of these terms correct

A1

All correct (no + c etc)

A1

3

(b) $x = -2 \quad \frac{dy}{dx} = 5 \times (-2)^4 + (40 \times -2)$

Substitute $x = -2$ into their $\frac{dy}{dx}$

M1

$\frac{dy}{dx} = 5 \times 16 + (40 \times -2) = 0 \Rightarrow P$ is stationary point

*CSO Shown = 0 plus statement
eg "st pt", "as required", "grad = 0" etc*

A1

Or their $\frac{dy}{dx} = 0 \Rightarrow x^a = k$

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(M1)

$x^3 = -8 \Rightarrow x = -2$

CSO $x = 0$ need not be considered

(A1)

2

(c) (i) $\frac{d^2y}{dx^2} = 20x^3 + 40$

Correct ft their $\frac{dy}{dx}$

B1

$$= 20 \times (-2)^3 + 40$$

Subst $x = -2$ into their second derivative

M1

$$(-160 + 40) = -120$$

CSO

A1

3

- (ii) Maximum (value)
 their c(i) answer must be < 0
 Other valid methods acceptable provided
 "maximum" is the conclusion

*Accept minimum if their c(i) answer > 0
 and correctly interpreted
 Parts (i) and (ii) may be combined by
 candidate but -120 must be seen to award
 A1 in part (c)(i)*

E1ft

1

(d) (When $x = 1$) $y = 13$

B1

When $x = 1$, $\frac{dy}{dx} = 5 + 40$

Sub $x = 1$ into their $\frac{dy}{dx}$

M1

$y = (\text{their } 45)x + k$ OE

ft their $\frac{dy}{dx}$

m1

Tangent has equation $y - 13 = 45(x - 1)$

CSO OE $y = 45x + c$, $c = -32$

A1

4

[13]

Q31.

- (a) When $x = 2$, $y = -3$

PI

B1

Use of $(2 + h)^2 = 4 + 4h + h^2$

M1

Correct method for gradient

M1

$$\text{Gradient} = \frac{-3 - 2h + h^2 + 3}{h} = -2 + h$$

A1 if only one small error made

A2, 1

5

(b) As h tends to 0,

E1 for 'h = 0'

E2,1

... the gradient tends to -2

*dependent on at least E1
ft small error in (a)*

B1F

3

[8]

Q32.

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(a) $\frac{dy}{dx} = 4x^3 - 32$

Reduce one power by 1

M1

One term correct

A1

All correct (no + c etc)

A1

3

(b) Stationary point $\Rightarrow \frac{dy}{dx} = 0$

M1

$$\Rightarrow x^3 = 8$$

$$x^n = k \text{ following from their } \frac{dy}{dx}$$

A1ft

$$\Rightarrow x = 2$$

CSO

A1

3

(c) (i) $\frac{d^2y}{dx^2} = 12x^2$

$$FT \text{ their } \frac{dy}{dx}$$

B1ft

1

(ii) When $x = 2$, $\frac{d^2y}{dx^2}$ considered

$$\text{Or complete test with } 2 \pm \varepsilon \text{ using } \frac{dy}{dx}$$

M1

$$\Rightarrow \text{minimum point}$$

E1ft

2

(d) Putting $x = 0$ into their $\frac{dy}{dx}$ ($= -32$)

M1

$$\frac{dy}{dx} < 0 \Rightarrow \text{decreasing}$$

$$\text{Allow "increasing" if their } \frac{dy}{dx} > 0$$

A1ft

2

[11]

Q33.

(a) (i) $(-1 + h)^3 = -1 + 3h - 3h^2 + h^3$
PI

B1

$y_B = (-1 + 3h - 3h^2 + h^3) + 1 - h + 1$
ft numerical error

B1F

$\dots = 1 + 2h - 3h^2 + h^3$
convincingly shown (AG)

B1

3

(ii) Subtraction of 1 and division by h

M1M1

Gradient of chord $= 2 - 3h + h^2$

A1

3

(iii) As $h \rightarrow 0$, $\text{gr}(\text{chord}) \rightarrow \text{gr}(\text{tgt}) = 2$
E0 if 'h = 0' used;
ft wrong value of p

E1B1F

2

(b) (i) $x^2 = -1 - \frac{1}{2} = -1.5$

ft wrong gradient

M1

A1F

2

(ii) Tangent at A drawn

M1

α and x_2 shown correctly
dep't only on the last M1

A1

2

[12]

Q34.

(a) $V = x(9 - 3x)^2$

*Attempt at V in terms of x (condone slip
when rearranging formula for $y = 9 - 3x$)
or $(9 - 3x)^2 = 81 - 54x + 9x^2$*

M1

$$V = x(81 - 54x + 9x^2)$$

$$= 81x - 54x^2 + 9x^3$$

AG; no errors in algebra

A1

2

(b) (i) $\frac{dV}{dx} = 81 - 108x + 27x^2$

One term correct

M1

Another correct

A1

All correct (no + c etc)

A1

$$= 27(x^2 - 4x + 3)$$

CSO; all algebra and differentiation correct

A1

4

(ii) $(x - 3)(x - 1)$ or $(27x - 81)(x - 1)$ etc
"Correct" factors or correct use of formula

M1

$$\Rightarrow x = 1, 3$$

*SC: B1, B1 For $x = 1, x = 3$ found by
inspection (provided no other values)*

A1

2

(c) $\frac{d^2V}{dx^2} = -108 + 54x$ (condone one slip)

ft their $\frac{dV}{dx}$ (may have cancelled 27 etc)

M1

CSO; all differentiation correct

A1

2

(d) (i) $x = 3 \Rightarrow \frac{d^2 V}{dx^2} = 54; \quad x = 1 \Rightarrow \frac{d^2 V}{dx^2} = -54$

*ft their $\frac{d^2 V}{dx^2}$ and their **two** x -values*

B1ft

1

(ii) $(x =) 1$ (gives maximum value)

Provided their $\frac{d^2 V}{dx^2} < 0$

E1

1

(iii) $V_{\max} = 36$
CAO

B1

1

[13]

Q35.

(a) (i) $2x^2 + 2xh + 4xh \quad (= 54)$
Attempt at surface area (one slip)

M1

$\Rightarrow x^2 + 3xh = 27$

AG CSO

A1

2

(ii) $h = \frac{27 - x^2}{3x}$ or $h = \frac{9}{x} - \frac{x}{3}$ etc

Any correct form

B1

1

(iii) $V = 2x^2h = 18x - \frac{2x^3}{3}$

AG (watch fudging) condone omission of brackets

B1

1

(b) (i) $\frac{dV}{dx} = 18 - 2x^2$

One term correct "their" V

M1

All correct unsimplified $18 - 6x^2/3$

A1

2

(ii) Sub $x = 3$ into their $\frac{dV}{dx}$

Or attempt to solve their $\frac{dV}{dx} = 0$

M1

Shown to equal 0 plus **statement** that this implies a stationary point if verifying

CSO Condone $x = \pm 3$ or $x = 3$ if solving

A1

2

(c) $\frac{d^2V}{dx^2} = -4x$

(= -12)

FT their $\frac{dV}{dx}$

B1 ✓

$\frac{d^2V}{dx^2} < 0$

at stationary point $\Rightarrow$ maximum

FT their second derivative conclusion

If "their" $\frac{d^2y}{dx^2} > 0 \Rightarrow$ minimum etc

E1 ✓



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